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3 JUNE 1964

SUPERSEDING
MIL-A-27434
10 MAY 1960

MILITARY SPECIFICATION

ADAPTERS, CONNECTOR, COAXIAL, RADIOFREQUENCY, BETWEEN SERIES, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 Scope. This specification covers the general requirements for weatherproof, between series, radio-frequency, coaxial connector adapters having a nominal impedance of 50 ohms. (See 6.1 and 6.5).

1.2 Classification.

1.2.1 *Type designation.* The type designation of connector adapters is derived from the AN nomenclature system specified in Standard MIL-STD-196, and shall be as specified. (See 3.1 and 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 —Plastic, Molding Material, Polyethylene, Low and Medium Density.
O-F-499 —Flux, Brazing, Silver Alloy, Low Melting Point.
QQ-B-626 —Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips).
QQ-B-613 —Brass, Leaded and Non-Leaded; Plate, Rolled-Bar, Sheet and Strip.
QQ-B-750 —Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire and Structural and Special Shaped Sections.
QQ-C-530 —Copper-Beryllium Alloy Bars, Rods, and Wire.
QQ-C-533 —Copper-Beryllium Alloy Strip.
QQ-S-365 —Silver Plating (Electrodeposited).
QQ-S-561 —Solder; Silver.
QQ-S-571 —Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form.
QQ-W-343 —Wire, Electrical and Nonelectrical, Copper (Uninsulated).
PPP-B-566 —Boxes, Folding, Paperboard.
PPP-B-585 —Boxes, Wood, Wirebound.
PPP-B-591 —Boxes, Fiberboard, Wood-Cleated.

FSC 5935

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PPP-B-601 —Boxes, Wood, Cleated-Plywood.
PPP-B-621 —Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 —Boxes, Fiberboard.
PPP-B-665 —Boxes: Paperboard, Metal-Stayed (Including Stay Material).
PPP-B-676 —Boxes, Set-Up, Paperboard.
PPP-T-76 —Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).
PPP-T-97 —Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-C-71 —Connectors, Coaxial, Radiofrequency, Series N, and Associated Fittings.
MIL-P-77 —Plastic Sheet and Plastic Rod, Thermosetting, Cast.
MIL-P-116 —Preservation, Methods of.
MIL-R-3065 —Rubber and Synthetic Rubber Compounds, General Purpose (Except Tires, Inner Tubes. Sponge Rubber, and Hard Rubber).
MIL-C-3608 —Connectors, Coaxial, Radio Frequency, Series BNC, and Associated Fittings, General Specification for.
MIL-C-3643 —Connectors, Coaxial, Radio Frequency, Series HN and Associated Fittings, General Specification for.
MIL-C-3650 —Connectors Coaxial, Radio Frequency, Series LC, General Specification for.
MIL-C-3655 —Connectors, Coaxial, Radio Frequency, Series Twin, and Associated Fittings, General Specification for.
MIL-C-3989 —Connectors, Coaxial, Radio Frequency, Series C, and Associated Fittings.
MIL-P-5516 —Packings and Gaskets, Preformed, Petroleum Hydraulic Fluid Resistant.
MIL-R-5847 —Rubber, Silicone; Low and High Temperature and Tear Resistant.
MIL-H-7199 —Heat Treatment of Beryllium Alloys, Process for.
MIL-B-10377 —Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 —Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-19468 —Plastic Rods, Polytetrafluoroethylene, Molded and Extruded.

DENTAL SPECIFICATIONS

(For applicable detail specifications, see Supplement 1.)

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-130 —Identification Marking of U. S Military Property.

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MIL-STD-196 —Joint Electronics Type Designation System.
MIL-STD-202 —Test Methods for Electronic and Electrical Component Parts.
MS33586 —Metal. Definition of Dissimilar.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Ave., at 33rd St., New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Detail requirements for individual connector adapter types. Detail requirements or exceptions applicable to individual types of connector adapters shall be as specified in the detail specifications listed in Supplement 1 of this specification. In the event of any conflict between requirements of this specification and the detail specifications, the latter shall govern. (See 6.2)

3.2 Material. Unless otherwise specified (see 3.1), the material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the connector adapters to meet the performance requirements of this specification. Acceptance or approval of any constitutent material shall not be construed as a guaranty of the acceptance of the finished product. Dissimilar metals between which an electromotive couple may exist shall not be placed in contact with each other. Reference is made to Standard MS-33586 for definition of dissimilar metals.

3.2.1 *Brass.* Brass shall conform to composition 11, half-hard, of Sp cification QQ-B-613, or composition 22, half-hard, of Specification QQ-B-626, as applicable.

3.2.2 *Copper beryllium.* Copper beryllium shall conform to condition H of Specification QQ-C-530 or QQ-C-533, as applicable. Parts fabricated of copper beryllium shall be heat-treated to condition HT after machining and forming. Heat treatment shall be in accordance with MIL-H-7199.

3.2.3 *Copper wire.* Copper wire shall conform to QQ-W-343.

3.2.4 *Phosphor bronze.* Phosphor bronze shall conform to composition A of Specification QQ-B-750.

3.2.5 *Dielectric materials.* Dielectric materials shall conform to the following:

3.2.5.1 *Polytetrafluoroethylene.* Polytetrafluoroethylene shall be in accordance with MIL-P-19468.

3.2.5.2 *Polyethylene.* Polyethylene shall conform to type II, grade 4, Class L of Specification L-P-390.

3.2.5.3 *Polystyrene.* Polystyrene shall conform to type E-2 of MIL-P-77.

3.2.6 *Silicone rubber.* Silicone rubber shall conform to class II, grade 50 or 60, of MIL-R-5847, except that the oil-immersion test is not applicable.

3.2.7 *Silver solder.* Silver solder shall conform to class 1 of Specification QQ-S-561.

3.2.7.1 *Flux.* Flux used while silver soldering shall conform to Specification O-F-499.

3.2.8 *Soft solder.* Soft solder shall conform to composition Sn60 of Specification QQ-S-571.

3.2.9 *Synthetic rubber.* Synthetic rubber shall conform to grade RS515BFF of MIL-R-3065.

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3.2.10 *Packing.* Packing shall conform to class B of MIL-P-5516.

3.3 **Design and construction.** Connector adapters shall be of the design, construction, and envelope dimensions specified. (See 3.1.) Wherever feasible, parts having similar electrical characteristics may be combined (fabricated as a single piece) to simplify construction. Parts of unlike materials may be combined such as brass and copper beryllium, provided copper beryllium is used in the fabrication of the single-piece construction.

3.3.1 *Terminations.* Unless otherwise specified herein, the requirements for the various terminations of the connector adapters shall conform to the specification specified in table I, as applicable. The adapters shall be gaged in accordance with 4.6.2.

TABLE I. *Requirements for terminations.*[1]

Termination requirements[2]	Specifications					
	MIL-C-3959	MIL-C-71	MIL-C-3608	MIL-C-3643	MIL-C-3650	MIL-C-3655
Series C						
Contacts						
Form-over solder joint						
Coupling sleeves:[3]						
Sleeve bushings						
Solder joint	X					
Rotation						
Interchangeability (applicable to male sections of adapters).						
Series N						
Coupling nuts						
Contacts		X				
Series BNC						
Coupling sleeves						
Contacts			X			
Series HN						
Coupling nuts						
Contacts				X		
Series LC						
Coupling nuts						
Contacts					X	
Series Twin						
Coupling nuts						
Contacts						X

[1] For the purpose of this specification, terminations are defined as the respective portions of the connector adapter which form the transition from one connector series to another.
[2] For series BN, LN, LT, SM, QDL, QDS, and UHF, see 3.1.
[3] Refers to bayonet-sleeve bushing and plain-sleeve bushing after assembly.

3.3.1.1 *Contacts.* When tested in accordance with 4.6.3, the contacts shall meet the dimensional requirements for contacts specified in the specifications specified in table I, as applicable. For series BN, LN, LT, SM, QDL, QDS, and UHF contacts, see 3.1.

3.3.2 *Finish.* Unless otherwise specified (see 3.1), all metal parts shall have a silver plating of not less than 0.0002 inch thick, and of sufficient smoothness and density to withstand the salt-spray (corrosion) test specified in 4.6.3. Silver plating shall be in accordance with Specification QQ-S-365. Dimensions of metal parts shall include the plating. (See 3.1).

3.3.3 *Screw threads.* Screw threads shall conform to Handbook H28, and shall have the specified fit after plating. (See 3.1.)

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3.4 Dielectric withstanding voltage. When connector adapters are tested as specified in 4.6.4, there shall be no evidence of breakdown

3.5 Salt spray (corrosion). When connector adapters are tested as specified in 4.6.5, there shall be no evidence of destructive corrosion or pitting. Destructive corrosion shall be construed as any type of corrosion which in any way interferes with mechanical or electrical performance.

3.6 Vibration. When adapters are tested as specified in 4.6.6, there shall be no evidence of mechanical damage, loosening of parts, rupturing, or electrical discontinuity exceeding 10 microseconds.

3.7 Shock. When adapters are tested as specified in 4.6.7, there shall be no evidence of mechanical damage, loosening of parts, rupturing, or electrical discontinuity exceeding 10 microseconds.

3.8 Marking. Connector adapters shall be marked in accordance with Standard MIL-STD-130, with the type designation and the manufacturer's code symbol. Marking shall be depressed in characters approximately 1/16 inch in height, in the place specified (See 3.1)

3.9 Workmanship. Connector adapters shall be processed in such a manner as to be uniform in quality and shall be free from sharp edges and burrs, except where sharp edges are required for mechanical or electrical reasons. All solder joints shall be thoroughly cleaned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier*. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 *Test equipment and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy. quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.1.1.2 *Additional inspection.* Nothing specified herein shall preclude the supplier from taking such additional samples and making such additional inspection as he may deem necessary or desirable to assure conformance of the connector adapters to this specification.

4.1.2 *Government* Acceptance of the connector adapters shall be based upon verification by the Government of the supplier's compliance with the requirements of this specification. The Government may, at its option, repeat any or all of the inspections specified herein. (See 6 3.)

4.2 Classification of inspection. The examination and testing of connector adapters shall be classified as follows:

(a) Component-materials inspection. (See 4.3.)

(b) Quality conformance inspection (See 4.5.)

1 Inspection of product for delivery **(See 4.5.1.)**

2. Inspection of preparation for delivery. (See 4 5.2.)

4.3 Component-materials inspection. Component-materials inspection shall consist of verification that the component materials listed in table II, used in fabricating the connector adapters, are in accordance with the applicable referenced specifications or requirements prior to such fabrication When deemed necessary by the Government certified test data or a certificate of analysis may be required from the source

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TABLE II. *Component-materials inspection*

Component material	Requirement paragraph	Applicable specification
Brass	3.2.1	QQ-B-613 or QQ-B-626
Copper beryllium	3.2.2	QQ-C-530 or QQ-C-533; and MIL-H-7199.
Copper wire	3.2.3	
Phosphor bronze	3.3.4	
Dielectric materials:		QQ-W-343
Polytetrafluoroethylene	3.2.5.1	QQ-B-750 MIL-P-19468
Polyethylene	3.2.5.2	L-P-390
Polystyrene	3.2.5.3	MIL-P-77
Silicone rubber	3.2.6	MIL-R-5874
Silver solder	3.2.7	QQ-S-561
Flux	3.2.7.1	O-F-499
Soft solder	3.2.8	QQ-S-571
Synthetic rubber	3.2.9	MIL-R-3065
Packing	3.2.10	MIL-P-5516

4.4 **Inspection conditions.** Unless otherwise specified herein, all inspection shall be made at room ambient temperature, relative humidity, and pressure.

4.5 **Quality conformance inspection.**

4.5.1 *Inspection of product for delivery.* Inspection of product for delivery shall consist of groups A, B, and C.

4.5.1.1 *Inspection lot.* An inspection lot, as far as practicable, shall consist of all the connector adapters of the same type designation, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 *Rejected lots.* If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.1.3 *Group A inspection.* Group A inspection shall consist of the examinations and test specified in table III, and shall be made on the same set of sample units, in the order shown.

4.5.1.3.1 *Sampling plan.* Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be as specified in table III. Major and minor defects shall be as defined in Standard MIL-STD-105.

4.5.1.4 *Group B inspection.* Group B inspection shall consist of the examinations and test specified in table IV

TABLE III. *Group A inspection*

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination:		4.6.1		
Design and construction	3.3		1.0	4.0
Finish	3.3.2			
Screw threads	3.3.3			
Material	3.3			
Marking	3.8			
Workmanship	3.2			
Electric withstanding voltage	3.4	4.6.2	1.0	

4.5.1.4.1 *Sampling plan.* The sampling plan shall be in accordance with Standard MIL-STD-105 for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. Small-sample reduced inspection procedure R-1 shall be used. The AQL and inspection levels shall be as specified in table IV.

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TABLE IV. *Group B inspection*

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	Inspection level	
				Normal and tightened inspection	Reduced inspection
Physical dimensions .. .. (Terminations)	3.3.1	4.6.2	4	S-4	S-3
Gage test for contacts .	3.1 and 3.3.1.1	4.6.3	4	S-4	S-3

4.5.1.4.2 *Disposition of sample units.* Sample units which have passed all the group B inspection, shall be delivered on the contract or order, if the lot is accepted.

4.5.1.4.3 *Action in case of lot rejection.* When an inspection lot is rejected on group B inspection, the supplier shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

4.5.1.5 *Group C inspection.* Group C inspection shall consist of the examinations and tests specified in table V.

TABLE V. *Group C inspection*

Examination or test	Requirement paragraph	Method paragraph
Subgroup 1 (unassembled connector adapters)		
Detail requirements:		
Physical dimensions [1]	3.1	4.6.1
Subgroup 2 (assembled connector adapters)		
Vibration	3.6	4.6.6
Shock	3.7	4.6.7
Salt spray (corrosion) . ..	3.5	4.6.5

[1] Only those dimensions related to piece parts shall be checked

4.5.1.5.1 *Sampling plan.*

4.5.1.5.1.1 *Subgroup 1 (unassembled connector adapters).* Six sample units of each type connector adapter shall be selected from initial production. Every 30 days thereafter, sample units shall be selected from subsequent production in accordance with table VI.

TABLE VI. *Sample size and acceptance number*

Number of connector adapters of a given type produced during a month [1]	Sample size	Acceptance number [2]	
		Subgroup	Subgroup
500 to 1,000, incl	2	0	0
1,001 to 5,000, incl .	6	1	0
5,001 to 15,000, incl	15	2	1

[1] When less than 500 connector adapters of a given type are produced in a month, the sample shall be taken from 2-month's production
[2] For subgroup 1, see 4.5.1.5.1.1 to 4.5.1.5.1.1.2, inclusive, for subgroup 2, see 4.5.1.5.1.2.

4.5.1.5.1.1.1 *Physical dimensions* To facilitate inspection of the physical dimensions, the unassembled sample units shall be divided into groups of identical piece parts. Inspection of the physical dimensions shall then be performed on a group-by-group basis

4.5.1.5.1.1.2 *Acceptance criteria* The acceptance criteria shall be determined as follows.

(a) For each group of identical piece parts, the number of defects in any one particular dimension (e.g., a flat washer would have 3 particular dimensions: inside diameter, outside diameter, and thickness) shall not exceed the applicable acceptance number specified in table VI, with a total of not more than 10 defects per

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100 dimensions for all piece parts in all groups combined.

4.5.1.5.1.2 *Subgroup 2 (assembled connector adapters).* Six sample units of each type connector adapter shall be selected from initial production. Every 30 days thereafter, sample units shall be selected from subsequent production in accordance with table VI. If the number of failures exceeds the acceptance number specified in table VI, the sample shall be considered to have failed.

4.5.1.5.1.2.1 *Disposition of sample units.* Sample units which have been subjected to group C inspection, subgroup 2, shall not be delivered on the contract or order.

4.5.1.5.2 *Noncompliance.* If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the contracting officer

4.5.2 *Inspection of preparation for delivery.* Sample items and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with requirements in section 5 of this specification.

4.6 Methods of examination and test.

4.6.1 *Visual and mechanical examination.* Connector adapters shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.1 to 3.3.3, incl, 3.8, and 3.9.)

4.6.2 *Terminations.* Connector adapters shall be examined by use of gages or other means to verify their conformance with table I and the dimensions given in figure 1 and figure 2. (See 3.3.1.)

4.6.3 *Gage test for contacts.* Contacts shall be gaged or measured to determine compliance with 3.3.1.1.

4.6.4 *Dielectric withstanding voltage (see 3.4).* Connector adapters shall be tested in accordance with method 301 of Standard MIL-STD-202. The following details shall apply:

(a) Special preparations or conditions:

1. The maximum relative humidity shall be 50 percent. When facilities are not available at this test condition, connector adapters shall be tested at room ambient relative humidity. In case of dispute, if the test has been made at room ambient relative humidity, retest shall be made at 50 percent maximum relative humidity.
2. Precautions shall be taken to prevent air-gap voltage breakdowns by use of a mating connector or a jig simulating a mating connector.
3. The voltage shall be metered on the high side of the transformer to an accuracy of 5 percent.
4. Connector adapters having different voltage requirements at each end shall be subjected to the lower voltage. (See 3.1.)

(b) Magnitude of test voltage—Unless otherwise specified (see 3.1), 1,500 volts root mean square instantaneously applied.

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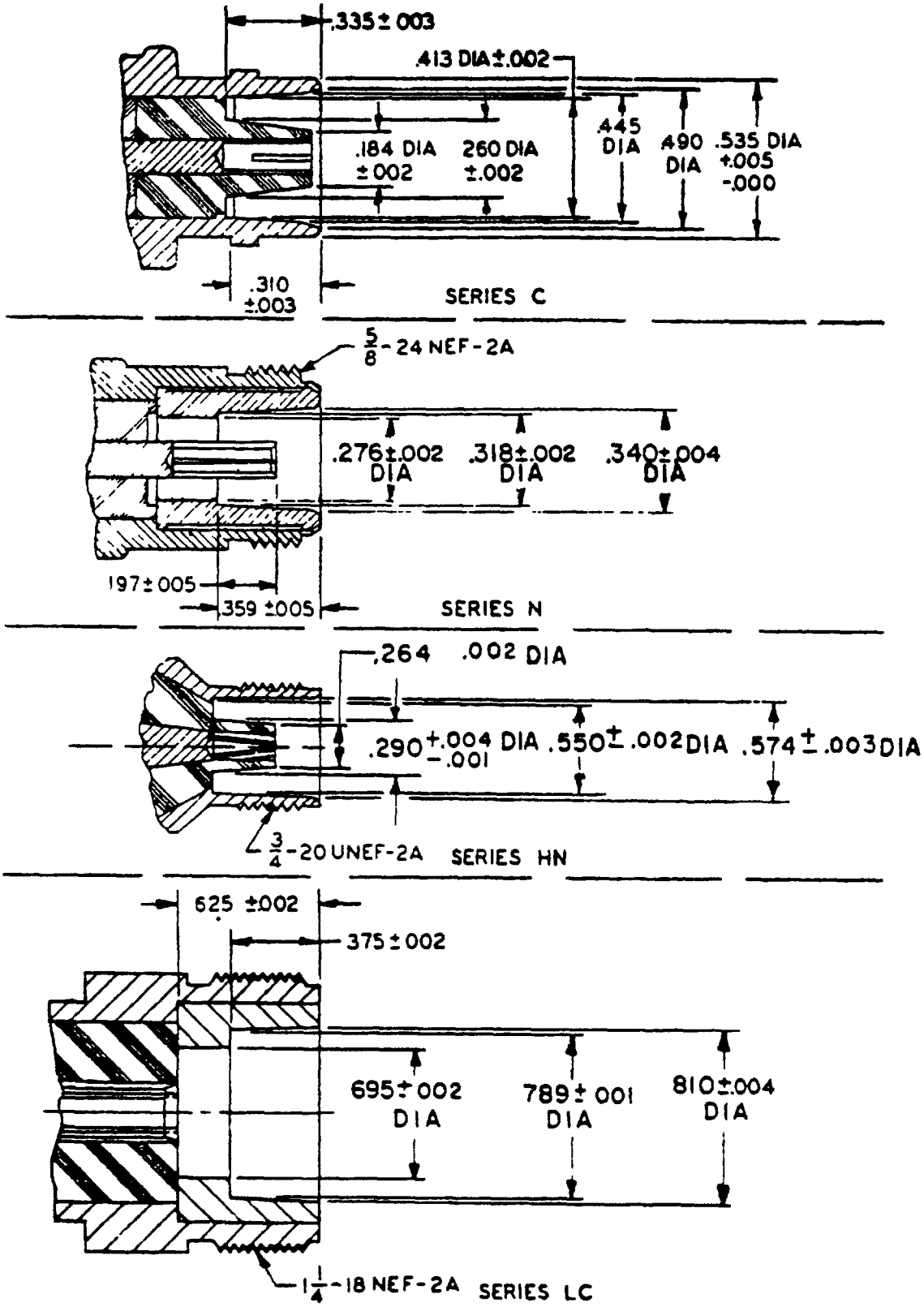
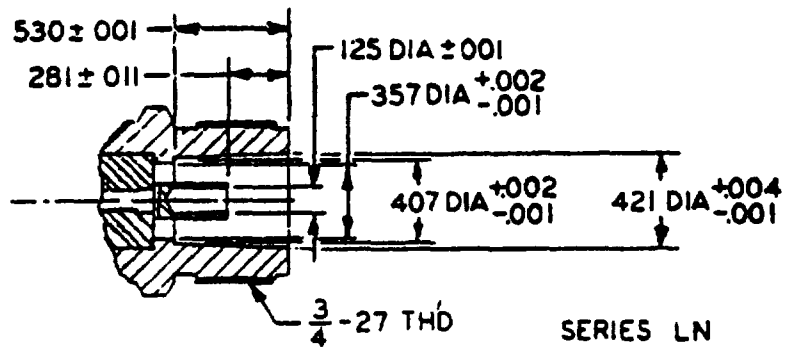
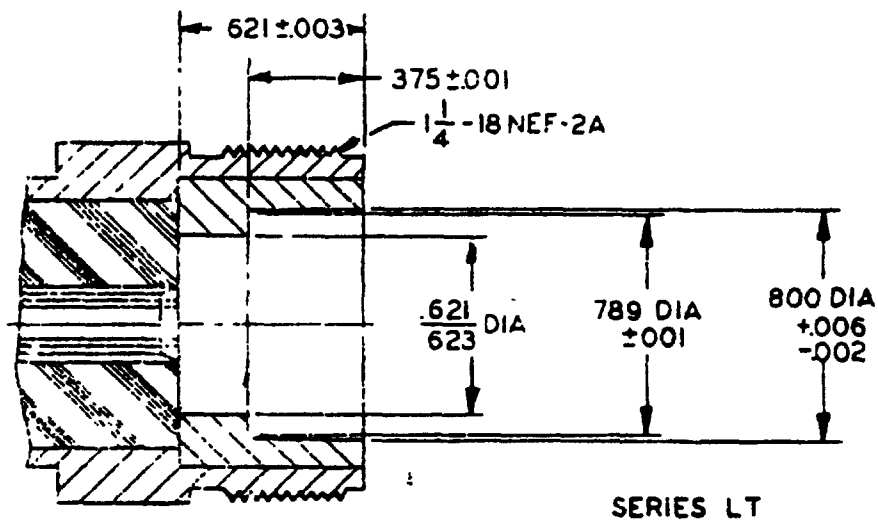
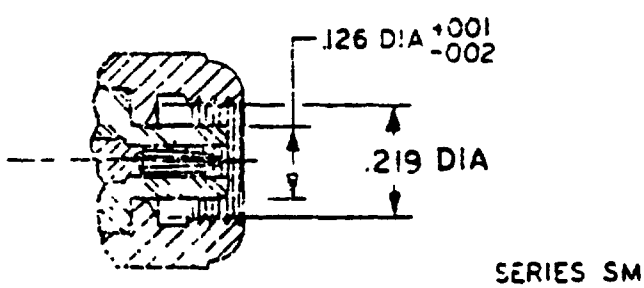


FIGURE 1. *Dimensional requirements for female terminations*

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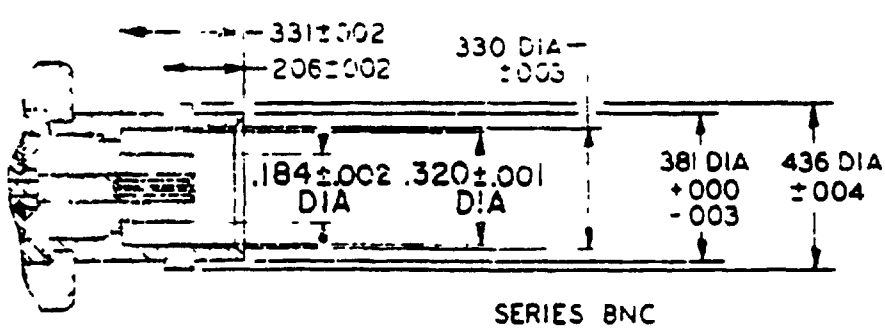
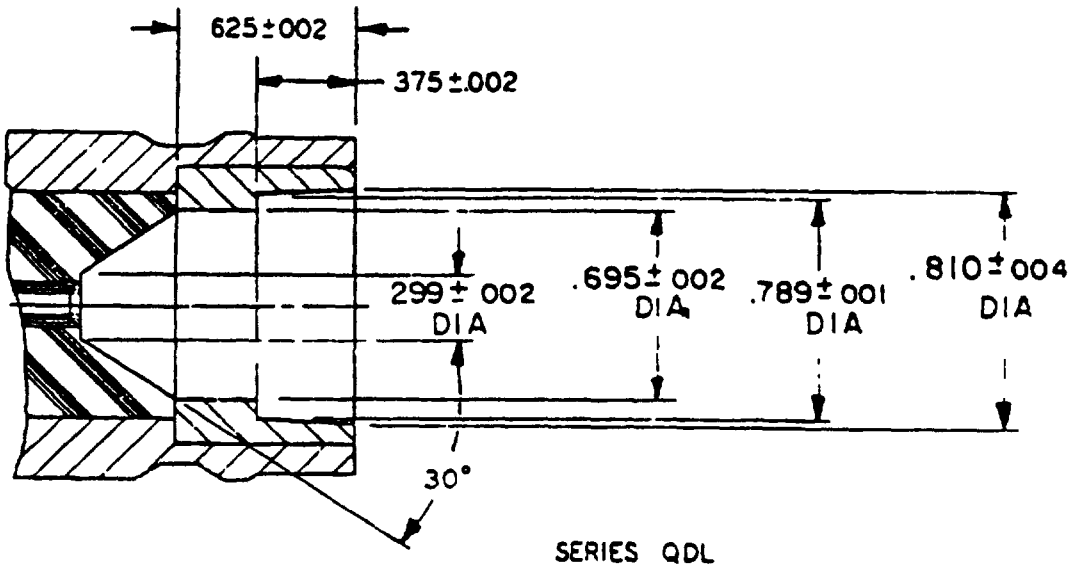
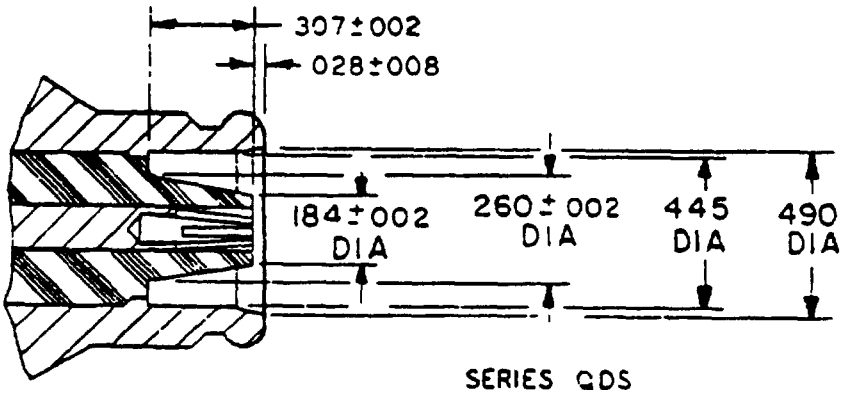


FIGURE 1 *Continued*

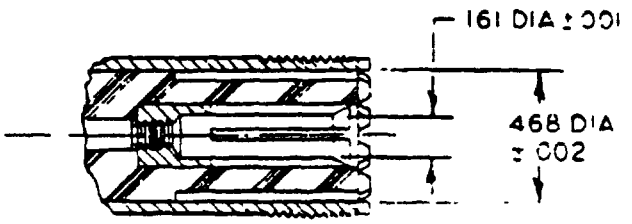
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SERIES QDL



SERIES QDS



SERIES UHF

NOTES.

1 Dimensions are in inches Unless otherwise specified tolerances on decimals are ±005

FIGURE 1 *Continued.*

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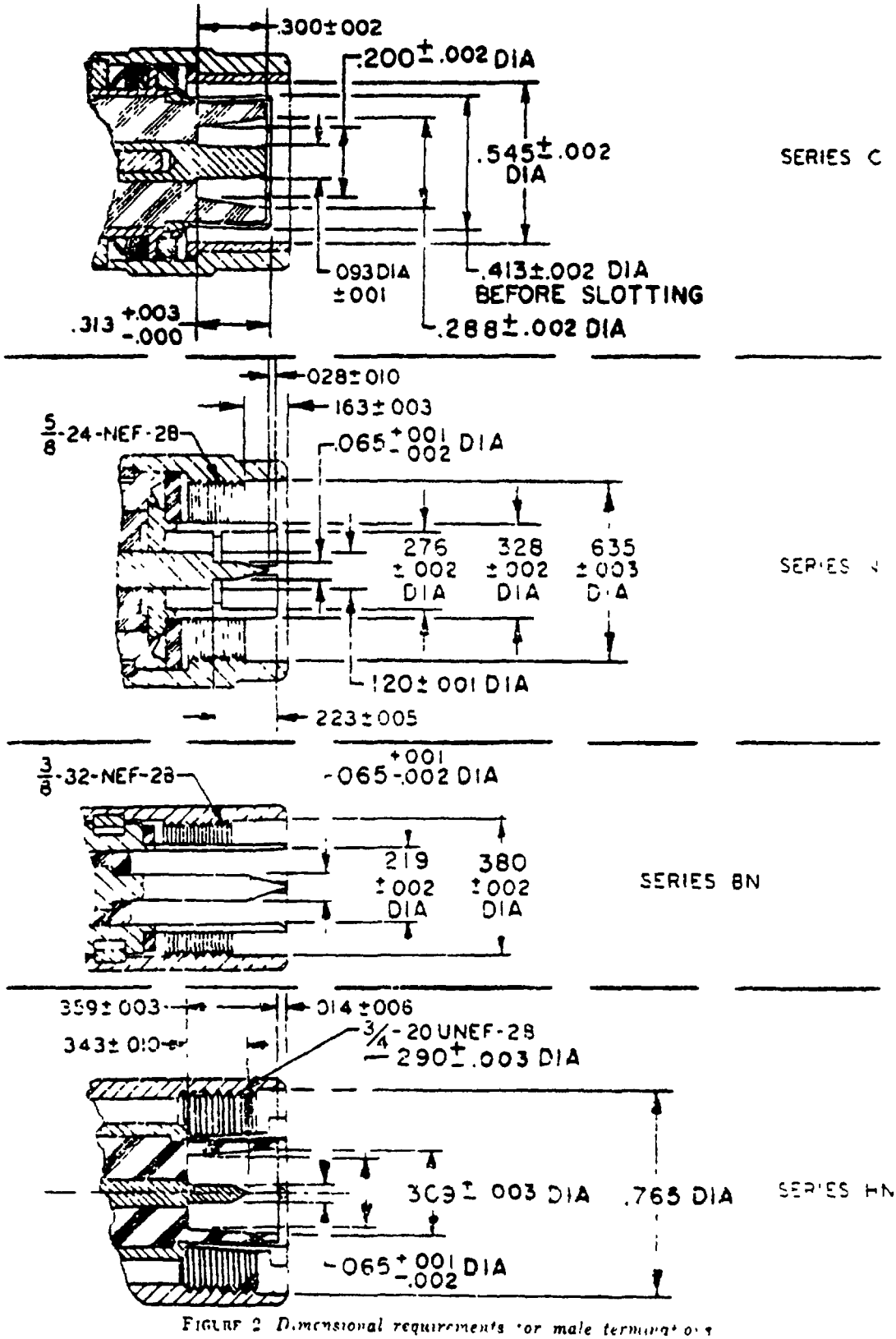
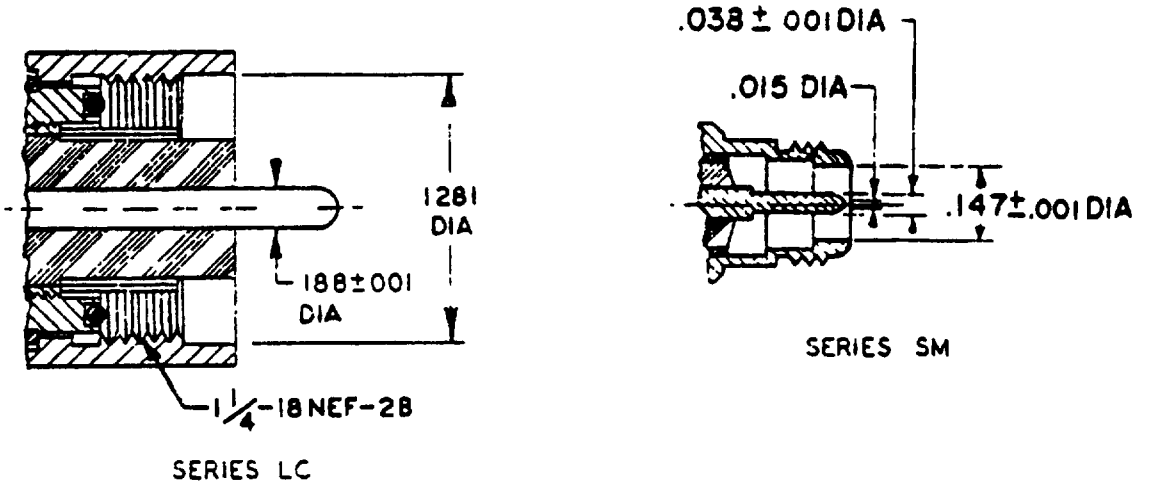
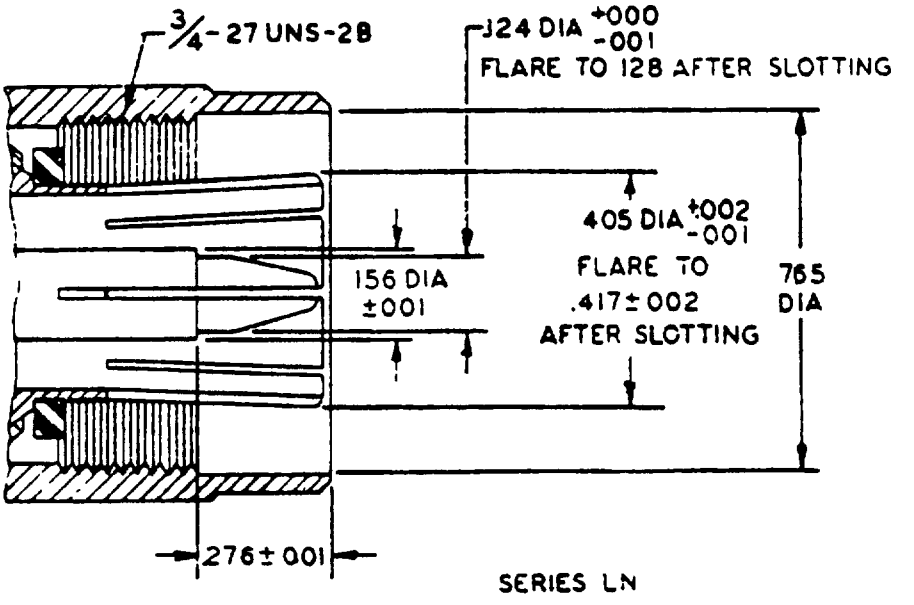


FIGURE 2. *Dimensional requirements for male terminations.*

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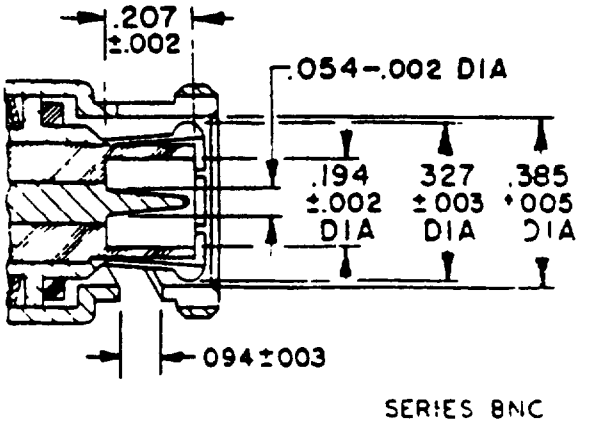
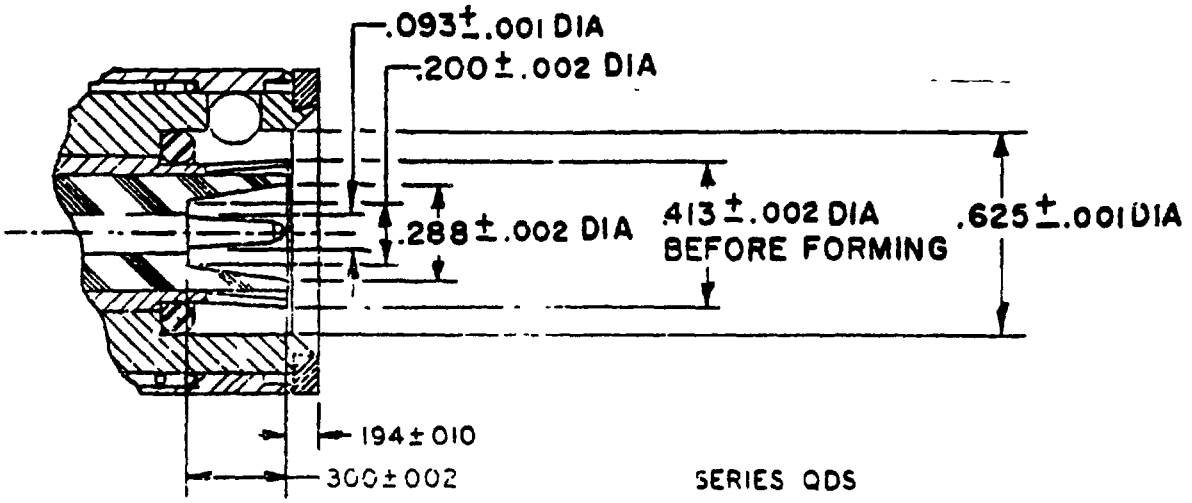
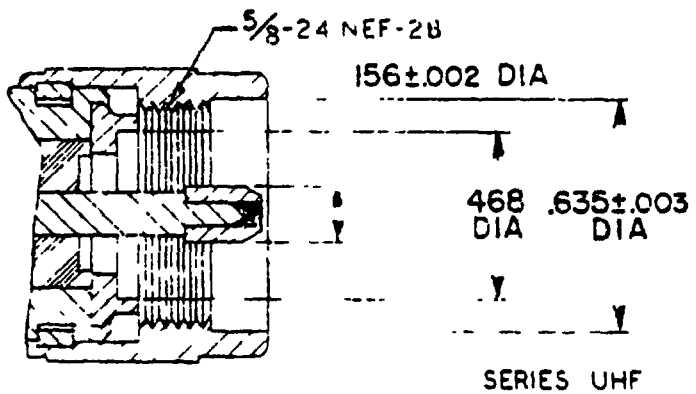
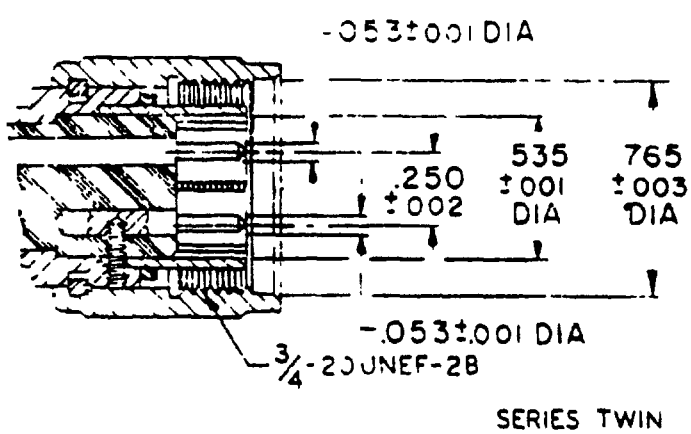


FIGURE 2 *Continued*

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NOTES
1 Dimensions are in inches Unless otherwise specified, tolerances on decimals are ± 005

FIGURE 2 *Continued*

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(c) Nature of potential—Alternating current.

(d) Points of application of test voltage—Between the center contact and body.

(e) Examination—After the test, connector adapters shall be examined for evidence of breakdown.

4.6.5 *Salt spray (corrosion).* Connector adapters shall be tested in accordance with method 101, test condition B, of Standard MIL-STD-202. At the conclusion of the test, the connector adapters shall be washed, shaken, air blasted, and then permitted to dry for 24 hours at 40° C. The connector adapters shall then be examined for evidence of corrosion. (See 3.5.)

4.6.6 *Vibration* (see 3.6). Adapters properly mated with their mating connectors shall be tested in accordance with method 204 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting of specimens—Adapters shall be mounted by normal mounting means.

(b) Electrical load conditions—No load.

(c) Test condition letter—B.

(d) Measurements—During vibration each adapter shall be monitored to determine electrical discontinuity.

4.6.7 *Shock.* Adapters properly mated with their mating connectors shall be tested in accordance with method 202 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting method—By normal mounting means.

(b) Acceleration requirements—100 G's.

(c) Number of blows—Adapters shall be subjected to nine blows, three in each of three mutually perpendicular planes.

(d) Electrical load—No load.

(e) Measurements—During shock each adapter shall be monitored to determine electrical discontinuity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

5.1.1 *Level A.* Unless otherwise specified (see 6.2), connector adapters shall be individually protected and unit-packaged in accordance with method IA of Specification MIL-P-P116, without the use of contact preservatives. Prior to packaging, the parts of the connector adapter shall be completely assembled. Unless otherwise specified (see 6.2), five unit packages or a multiple thereof shall be further packaged in intermediate containers conforming to Specifications PPP-B-566, PPP-B-665, or PPP-B-676. The gross weight of the intermediate container shall not exceed 10 pounds.

5.1.2 *Level C* Connector adapters shall be afforded preservation and packaging in accordance with the supplier's normal commercial practice.

5.2 Packaging (see 6.2).

5.2.1 *Level A.* Connector adapters packaged as specified (see 6.2) shall be packed in over-seas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood nailed wood, fiber (class 2 or 3, as specifie (see 6.2)), or wood-cleated paper-overlai boxes conforming to Specification PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636. and MIL-B-10377, respectively, at the option of the supplier. Shipping containers shall have case liners conforming to Specification MIL-L-10547; the case liners shall be closed and sealed in accordance with the appendix thereto Case liners for boxes conforming to Specification PPP-B-636 may be omitted provided the center and edge seams and manufacturers' joints are sealed with tape, at least 1½ inches wide, conforming to Specification PPP-T-76 Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to Specification PPP-B-636 may be banded with tape

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conforming to type IV of Specification PPP-T-97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.2 *Level B.* Connector adapters packaged as specified (see 6.2) shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 1 or 2, as specified (see 6.2)), or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, and MIL-B-10377, respectively, at the option of the supplier. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C* Connector adapters packaged as specified (see 6.2) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.4 *General.* Insofar as possible and practical, exterior containers shall be uniform in shape and size, shall be minimum cube and tare consistent with the protection required, and shall contain identical quantities of identical items.

5.3 **Marking.** In addition to any special marking required by the contract or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. (See 6.2.)

6. NOTES

6.1 **Intended use.** Connector adapters covered by this specification are intended for use in radio-frequency applications up to 10,000 megacycles per second. Their use is governed by temperature limitation of materials, and they are not recommended for use in applications where temperatures exceed 125° C.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable detail specification, and the complete type designation. (See 1.2.1 and 3.1.)

(c) That the supplier shall not substitute for a specified material or combination of fabricated parts (see 3.3) unless he obtains approval from the Government. Evidence to substantiate his claim that such a substitution is suitable shall be submitted with his request. Similar notification and substantiating evidence shall be submitted at any later time if substitution becomes necessary or desirable. At the discretion of the Government, test samples may be required to prove the suitability of the proposed substitute

(d) Levels of preservation and packaging and packing, and applicable marking (see section 5.)

(e) Method of preservation of packaging of Specification MIL-P-116, if other than method IA (See 5.1.1.)

(f) Number of unit packages if other than that specified in 5.1.1.

(g) Class of fiber. (See 5.2.1 and 5.2.2.)

6.3 **Government verification inspection.** Verification inspection by the Government will be limited to the amount deemed necessary to determine compliance with the contract or order and will be limited in severity

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to the definitive quality assurance provisions established in this specification and the contract or order. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the suppliers quality-control system and the quality history of the product. (See 4.1.2.)

6.4 Cross index. This specification covers the design of between-series connector adapters which were previously covered by single Service drawings and sheet form MS military standards. Table VII is a cross index of the connector adapters and single Service drawings and MS military standards.

TABLE VII *Cross index of connector adapters and Service drawings and MS military standards.*

Type designation	Bureau of Ships drawing	Signal Corps drawing	Air Force drawing	MS military standard
UG-108A/U	REB49277			
UG-201A/U	REB49324			
UG-218A/U	REB49274			
UG-217A/U	REB49203			
UG-218A/U	REB49275			
UG-259A/U	REB49204			
UG-270A/U	REB49278			
UG-271A/U	REB49279			
UG-327A/U	REB49287			
UG-335/U	REB49103			
UG-497A/U	REB49273			
UG-516A/U			51B12903	
UG-559B/U	REB49344		50C13669	
UG-564/U	REB49247			MS35321 (SigC)
UG-565A/U	REB49250			MS35324 (SigC)
UG-586B/U	REB49164			
UG-587B/U	REB49165			
UG-605/U	REB49271			
UG-606/U			51C12929	
UG-635/U	REB49237			MS35331 (SigC)
UG-636A/U	REB49238			MS35283 (SigC)
UG-637/U				MS35382 (SigC)
UG-690/U				MS35113 (SigC
UG-691/U				MS35114 (SigC
UG-702/U				MS90253
UG-703A/U				MS90261
UG-966/U	REB49121			
UG-970/U		SC-C-106729		
UG-971/U		SC-C-106728		
UG-977C/U	REB49126			
UG-1013A/U	REB49129			
UG-1034/U		SC-C-19743		
UG-1136/U	REB49306			
UG-1137B/U	REB49309			
UG-1142/U	REB49319			
UG-1143/U	REB49320			
UG-1144/U	REB49321			
UG-1146/U	REB49323			
UG-1306/U		SC-C-45985		

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6.5 Engineering information. Illustrations and additional engineering information for these connector adapters are available in the Military Standardization Handbook MIL-HDBK-216, R. F. Transmission Lines & Fittings, copies of which are available upon request from the Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.

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Navy—Ships
Air Force—Code 11

Reviewers
Army—ML, MI
Navy—ESO
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Users:
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Navy—MC, WEPS
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Preparing activity:
Army—EL

Project 5935-1077

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